

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.linear_model import LinearRegression
4 import joblib
5
6 data = {
7     'Advertising': [10, 20, 30, 40, 50, 60, 70, 80],
8     'Sales': [100, 150, 200, 250, 300, 350, 400, 450]
9 }
10 df = pd.DataFrame(data)
11
12 print("Dataset:")
13 print(df)
14
15 x = df[['Advertising']]
16 y = df['Sales']
17
18 model = LinearRegression()
19 model.fit(x, y)
20
21 print("\nModel Trained Successfully")
22 joblib.dump(model, "sales_model.pkl")
23 print("Model Saved Successfully")
24 loaded_model = joblib.load("sales_model.pkl")
25
26 new_data = [[90]]
27 prediction = loaded_model.predict(new_data)
28
29 print("\nPredicted Sales for Advertising = 90:")
30 print(prediction[0])
31 plt.plot(df['Advertising'], df['Sales'], marker='o')
32 plt.title("Advertising vs Sales")
33 plt.xlabel("Advertising Budget")
34 plt.ylabel("Sales")
35 plt.grid(True)
36 plt.show()
```

dataset.

	Advertising	Sales
0	10	100
1	20	150
2	30	200
3	40	250
4	50	300
5	60	350
6	70	400
7	80	450

Model Trained Successfully

Model Saved Successfully

C:\Users\DHARSHINI P\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names

warnings.warn(

Predicted Sales for Advertising = 90:

500.0

Advertising vs Sales

